

Two-Stage YOLO26 Instance Segmentation of Visible Post-Harvest Surface Defects on Apple and Tomato

Ralph Kevin G. Morales

*School of Information Technology
Mapúa University
Manila, Philippines
email@example.com*

Alexander Jann P. Espia

*School of Information Technology
Mapúa University
Manila, Philippines
email@example.com*

Deangelo James P. Largueza

*School of Information Technology
Mapúa University
Manila, Philippines
email@example.com*

Niel Francis M. Arligue

*School of Information Technology
Mapúa University
Manila, Philippines
email@example.com*

Lysa V. Comia

*School of Information Technology
Mapúa University
Manila, Philippines
email@example.com*

Abstract: Post-harvest grading datasets label whole images but do not show where a defect is. This study builds pixel-level masks for visible surface defects on apple and tomato and trains a two-stage YOLO26 instance segmentation model. Stage 1 learns the whole fruit. Stage 2 starts from the Stage 1 weights and learns three defect classes: bruise or discoloration, rot mold or decay, and surface damage. The apple set has 2,110 training files from 726 capture groups and frozen validation and test splits with no shared capture groups. On validation the YOLO26l Stage 1 model reached 98.5% mask mAP50 for the whole apple and the Stage 2 model reached 68.8% mask mAP50 over the three defect classes. On the untouched test split the Stage 2 model reached **[TBD]** mask mAP50. Tomato results are **[TBD]**. Surface damage was the hardest class. A Streamlit application runs both stages on uploaded photos.

Keywords: instance segmentation, post-harvest fruit defects, YOLO26, two-stage training, apple, tomato

Earlier studies show that pixel-level analysis helps produce inspection. Agarla et al. segmented apple defects using color and multispectral images [3]. Knott et al. combined banana foreground segmentation with individual defect masks for defect-based grading [4]. Leiva et al. trained YOLO11 in two phases on apples with scab, first on the fruit and then on the disease, and reported that the first phase helped the second [5]. Their results cannot be assumed to transfer to a new mixed-source dataset with different defect categories.

This study segments visible post-harvest surface defects on apple and tomato in RGB photos. It adapts the two-phase idea of [5] to YOLO26. Stage 1 segments the whole fruit. Stage 2 starts from the Stage 1 weights and segments three defect classes. The classes describe visible appearance only. They do not confirm a disease or internal damage.

The contributions are: (1) a reviewed set of polygon masks for three visible defect classes built on top of image-level grading photos, (2) a frozen leakage-controlled split by capture group, (3) a record of 20 training runs from the first pilot to the final model, (4) a two-stage YOLO26l model evaluated on an untouched test split, and (5) an upload-and-predict web application.

I. INTRODUCTION

Visible surface defects are an important part of post-harvest fruit quality assessment. A grade assigned to a whole image does not show where a defect is or how much of the surface it covers. AFruitDB provides 3,167 images of six fruit types sorted into three quality grades [1]. These image-level labels support grading research but do not give the pixel-level information needed to outline damaged regions.

Computer vision tasks differ in what they output. Image classification gives one label per image. Object detection draws a bounding box around each object. Instance segmentation outputs a mask for each object, which is the exact set of pixels it covers. Mask R-CNN established this approach by predicting a mask together with each class and box [2]. A box around a small or irregular bruise also contains healthy skin so a mask is a better fit for fruit inspection.

II. REVIEW OF RELATED LITERATURE

A. Fruit Grading Data and Pixel-Level Annotations

Mojumdar et al. introduced AFruitDB for grading six common Asian fruits [1]. Images are organized by fruit and grade. A grade does not mark the boundary of a bruise or scar

and a lower grade does not name a defect type. In this study defect labels were therefore assigned by inspecting each image and not inferred from grade folders.

Whole-fruit masks also change the role of healthy-looking images. Each image still needs a fruit mask and simply has no defect mask when careful inspection finds none. This keeps fruit identity separate from visible damage and allows several defect regions on one fruit.

B. Segmentation for Fruit Defect Assessment

Agarla et al. built a U-shaped network for quasi real-time apple defect segmentation and studied the value of near-infrared input [3]. Their work labels defective pixels rather than a whole-fruit grade. It used a different imaging setup and did not separate several visually similar defect types.

Tan et al. measured internal bruising in cut blueberries by segmenting the cross-section and the bruised tissue separately [6]. This is a useful precedent for pairing an object mask with a damage mask. It studied internal tissue so its scores do not apply to external defects on whole fruit.

Knott et al. used weakly supervised panoptic segmentation for banana grading with defect instances such as bruises and scars [4]. Box prompts and human checks supported mask preparation. Leiva et al. used YOLO11 segmentation for apple scab with a fruit phase before a disease phase [5]. The present study follows this staged design but uses three appearance classes and two fruits.

C. Instance Segmentation and Human-Reviewed Masks

Mask R-CNN shows that recognition and mask prediction can be learned in one framework [2]. A strong architecture still depends on consistent labels. Annotation quality is therefore treated as part of the research design.

The Segment Anything Model (SAM) produces masks from point or box prompts [7]. It can speed up outlining but a generated shape does not confirm that a region is a bruise or rot. Every assisted mask in this study was checked by a team member for both boundary and class.

D. Robustness and Evaluation

Nesteruk et al. combined image collection with augmentation for apple rot segmentation under changing conditions [8]. Their work shows how hard it is to transfer a model beyond its training conditions. Augmentation adds variety but does not replace evaluation on genuinely new

images. This study keeps a frozen test split that was never used for training or model selection.

E. Synthesis

The reviewed work supports three design choices: masks for irregular damage, a fruit mask kept separate from defect masks, and human review of assisted labels. It also shows that results on one fruit or setup do not prove performance on another. The question here is how well a two-stage YOLO26 model segments visible defects once grading photos are extended with reviewed masks.

III. DATASET AND ANNOTATION

A. Image Sources

Apple images come from three sources. AFruitDB grading photos [1] supply 986 training files. The Lab2Wild apple rot set by Nesteruk [9] supplies 843. Photos taken by the team supply 281. The original AFruitDB grade folders are image-level labels and were not used as training targets. All pixel masks were drawn by the team. Tomato images come from AFruitDB [1] and team captures. Tomato counts are reported in Section V.

B. Classes and Annotation Protocol

Polygons were drawn in Roboflow [10] and a local SAM-assisted tool [7] that proposes a mask from clicks. Every proposal was reviewed and corrected by a team member. Pilots began with five defect classes. Cut or crack and spot or scar were rare and often confused so they were combined as surface damage. Visible mold was combined with rot. The final classes are listed in Table I. Shadows, glare, stems, natural speckling and normal ripening color are excluded.

TABLE I. DEFECT CLASSES AND APPLE TRAINING COUNTS

Class	Visible appearance	Train images	Train masks
bruise_discoloration	Smooth dark or discolored patch	607	709
rot_mold_decay	Soft collapsed or moldy tissue	620	697
surface_damage	Cut crack scar or dry spot	547	873

C. Splits and Leakage Control

The Roboflow training export contains up to three augmented variants of many photos. The 2,110 training files therefore come from 726 capture groups and not 2,110 distinct photos. Each group stays in one split. Validation has 215 images from 210 groups and test has 79 images from 75 groups. A further 63 images form a protected reserve that was not used at all. Exact pixel hashes and capture groups were checked for overlap across splits and none was found. 545 training images have no defect mask and act as negatives.

IV. METHODOLOGY

A. Two-Stage Training

YOLO26 is the 2026 release of the Ultralytics YOLO family [11]. The large segmentation variant YOLO26l-seg was used. In Stage 1 the model learns a single fruit class on every training image including damaged fruit. In Stage 2 the Stage 1 best checkpoint initializes a new run with three defect outputs, a fresh AdamW optimizer and a new learning-rate schedule. Shared backbone and neck weights carry over and the output layers are rebuilt. Table II lists the settings.

TABLE II. TRAINING SETTINGS (APPLE)

Setting	Stage 1 (Run 21)	Stage 2 (Run 22)
Initial weights	yolo26l-seg.pt	Run 21 best.pt
Classes	apple	3 defects
Optimizer / lr0	SGD / 0.001	AdamW / 0.0005
Max epochs / patience	100 / 10	300 / 30
Image size / batch	864 / 56	864 / 56
LR schedule	cosine lrf 0.01	cosine lrf 0.01
Weight decay / warmup	0.0005 / 3	0.0005 / 3

Online augmentation was applied to training images only: horizontal and vertical flips ($p = 0.5$), rotation up to 10 degrees, translation 0.05, scale 0.1 and small hue saturation and value shifts (0.01, 0.2, 0.2). Mosaic, MixUp and Copy-Paste were off so that defect color and texture stay realistic. Training ran on one NVIDIA A100 80 GB GPU in Google Colab with Ultralytics 8.4.126, mixed precision and seed 0.

B. Evaluation Metrics

All scores are mask metrics. Precision is the share of predicted masks that match a labeled mask. Recall is the share

of labeled masks that are found. A match needs an intersection over union (IoU) above a threshold, where IoU is the overlap area divided by the combined area. mAP50 is the mean average precision at IoU 0.50 and mAP50-95 averages it over IoU 0.50 to 0.95. The best checkpoint was chosen on validation only. The test split was evaluated once after every choice was frozen.

V. RESULTS AND DISCUSSION

A. Run History

Twenty runs were recorded from 13 September to 1 October 2026 (Table III). Early YOLO26n pilots on five classes stayed below 0.50 mask mAP50. Merging rare classes and adding more reviewed images raised scores. Joint apple and tomato models reached about 0.75 overall mAP50 but most of that came from the easy whole-fruit classes. Runs on different datasets are historical and not controlled comparisons.

TABLE III. SELECTED RUNS (VALIDATION MASK METRICS)

Run	Data	Model	mAP50	mAP50-95
P1 pilot	Apple 5 cls	26n	0.474	0.310
P6 pilot	Apple+tomato	26s	0.687	0.506
Run 11	Apple+tomato	26m	0.715	0.598
Run 14	Apple 4 cls	26m	0.714	0.558
Run 20 S2	Apple defects	26m	0.671	0.450
Run 21 S1	Apple fruit	26l	0.985	0.985
Run 22 S2	Apple defects	26l	0.688	0.458

Runs 14, 20 and 22 share the same images and splits. The single-stage Run 14 averaged 0.624 mask mAP50 over its three defect classes. The two-stage YOLO26m Run 20 reached 0.671 and the two-stage YOLO26l Run 22 reached 0.688. These runs also differ in image size, optimizer and model size so the gain cannot be credited to the two-stage design alone.

B. Final Apple Model

Stage 1 segments the whole apple almost perfectly on validation with 98.5% mask mAP50 and 98.6% recall. Stage 2 reached 78.5% precision, 64.5% recall, 68.8% mAP50 and

45.8% mAP50-95 on validation at its best epoch. Table IV gives per-class results.

TABLE IV. APPLE STAGE 2 PER-CLASS MASK RESULTS

Class	Val P	Val R	Val mAP50	Test mAP50
Bruise	[TBD]	[TBD]	[TBD]	[TBD]
Rot	[TBD]	[TBD]	[TBD]	[TBD]
Surface	[TBD]	[TBD]	[TBD]	[TBD]
All	0.785	0.645	0.688	[TBD]

Figure pending: [TBD]

Fig. 1. Normalized validation confusion matrix of the apple Stage 2 model.

C. Unseen Test

The frozen test split (79 images, 75 capture groups) was evaluated once with the final Stage 2 checkpoint. Test mask precision was [TBD], recall [TBD], mAP50 [TBD] and mAP50-95 [TBD]. [TBD]

D. Tomato

Tomato follows the same two-stage recipe. [TBD] An earlier joint apple and tomato model (Run 11) reached 0.955 mask mAP50 for the whole tomato but only 0.419, 0.597 and 0.347 for tomato bruise, rot and surface damage on validation. Tomato defects were the weakest part of the project.

E. Errors and Limits

Surface damage is the hardest class because it covers small scars, cuts and dry spots with weak contrast. Most errors are missed small defects, confusion between bruise and early rot, and false positives on natural speckles or glare. The images come from a few sources with mostly plain backgrounds. Performance on new lighting, cultivars or conveyor images is unknown. Classes describe appearance only and the model does not diagnose disease or internal damage.

F. Web Application

A Streamlit application loads the Stage 1 and Stage 2 checkpoints for the selected fruit. Users upload a photo or pick a sample. The app overlays colored defect masks with a legend and reports each defect class with its share of the image and of the Stage 1 fruit mask. A confidence slider, class toggles, inference time and a PNG download are

included. Unreadable files and images with no defect produce clear messages.

VI. CONCLUSION AND FUTURE WORK

A two-stage YOLO26l model trained on team-reviewed masks segments the whole apple almost perfectly and segments three visible defect classes with 68.8% validation mask mAP50 and [TBD] test mask mAP50. Rot was the easiest defect and surface damage the hardest. Future work should add more reviewed small-defect examples, run a controlled single-stage baseline with identical settings, test on images from new sources and measure agreement between annotators.

REFERENCES

- [1] M. U. Mojumdar et al., "AFruitDB: A comprehensive dataset of six commonly used Asian fruits for advanced grading and biodiversity insights," *Data in Brief*, vol. 59, Art. no. 111380, 2025, doi: 10.1016/j.dib.2025.111380.
- [2] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2961–2969.
- [3] M. Agarla, P. Napoletano, and R. Schettini, "Quasi real-time apple defect segmentation using deep learning," *Sensors*, vol. 23, no. 18, Art. no. 7893, 2023, doi: 10.3390/s23187893.
- [4] M. Knott, D. Odion, S. Sontakke, A. Karwa, and T. Defraeye, "Weakly supervised panoptic segmentation for defect-based grading of fresh produce," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2025.
- [5] F. Leiva, S. G. Rebeaud, and D. Christen, "Real-time identification and quantification of apple scab on fruit in preharvest and postharvest conditions using YOLO11: a deep learning approach," *Plant Methods*, vol. 22, Art. no. 29, 2026, doi: 10.1186/s13007-026-01508-7.
- [6] C. Tan, C. Li, P. Perkins-Veazie, H. Oh, R. Xu, and M. Iorizzo, "High throughput assessment of blueberry fruit internal bruising using deep learning models," *Front. Plant Sci.*, vol. 16, Art. no. 1575038, 2025, doi: 10.3389/fpls.2025.1575038.
- [7] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 4015–4026.
- [8] S. Nesteruk et al., "XtremeAugment: Getting more from your data through combination of image collection and image augmentation," *IEEE Access*, vol. 10, pp. 24010–24028, 2022, doi: 10.1109/ACCESS.2022.3154709.
- [9] S. Nesteruk, "Apple rotting segmentation problem in the wild," *Kaggle dataset*, CC BY-NC-SA 4.0. [Online]. Available: <https://www.kaggle.com/datasets/sergeynesteruk/apple-rotting-segmentation-problem-in-the-wild>
- [10] Roboflow, "Roboflow Annotate." [Online]. Available: <https://roboflow.com>
- [11] Ultralytics, "YOLO26," *Ultralytics documentation*, 2026. [Online]. Available: <https://docs.ultralytics.com/models/yolo26/>

